REAL-TIME BOLT AND NUT MOUNTING VERIFICATION SYSTEM USING COMPUTER VISION

BE23PW701 & BE23PW702

MAKE – A – PRODUCT - PHASE I & II

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BONAFIDE CERTIFICATE

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE. NO
1	**BUSINESS IDEA AND PROBLEM STATEMENT (STAGE -I)** 1.1 Major Domain 1.2 Target Customers 1.3 Go To Customer (GTC) Process 1.4 Potential Problem received 1.5 Specific Problem chosen & Rationale - Customer Fit 1.6 Final Problem Statement 1.7 GTC and Problem – Customer Fit	**3 - 4 Pages**
2	**DESIGN AND DEVELOPMENT OF SOLUTION (STAGE -II)** 2.1 Solution Process / Methodology 2.2 Minimum Viable Product (MVP) 2.3 Solution – Customer Fit 2.4 Concept MVP	**3 – 4 Pages**
3	**PROTOTYPE DEVELOPMENT (STAGE -III)** 3.1 Prototype Development 3.2 GTC Iteration & Product Improvement 3.3 Product – Customer Fit	**3 - 4 Pages**
4	**BUSINESS MODEL DEVELOPMENT (STAGE -IV)** 4.1 Lean Business Model Canvas (BMC) 4.2 Market Potential 4.3 Business Plan and Marketing plan 4.4 Business – Customer Fit 4.5 Patenting Potential and Startup Potential 4.6 SDG Alignment 4.7 Unique Selling Proposition (USP)	**4 - 5 Pages**
5	**SUMMARY (STAGE V)** 5.1 Minimum Viable Product (MVP) 5.2 USP (Technical Dimension) 5.3 USP (Business Dimension) 5.4 Innovativeness & IP Potential 5.5 Scope for Startup Creations 5.6 Final Minimum Viable Product (MVP)	**2 Pages**
6	**KEY LEARNING OF THE TEAM (10 POINTS) (STAGE -VI)**	**1 Page**
7	**REFERENCES**	**1 - 2 Pages**

1. BUSINESS IDEA AND PROBLEM STATEMENT (STAGE -I)

1.1 MAJOR DOMAIN

The proposed project is primarily classified under the domains of Industrial Automation, Computer Vision, and Smart Manufacturing Systems. In modern manufacturing environments, there is an increasing shift toward automation to improve productivity, reduce human intervention, and ensure consistent product quality. This project contributes to that transition by developing a vision-based inspection system for fastener verification.

From a technical perspective, the system integrates multiple engineering disciplines:

- Mechanical Engineering: Design and implementation of the conveyor system for controlled object movement
- Electronics and Control Systems: Use of IR sensor for object detection and synchronization
- Computer Science / Software Engineering: Implementation of image processing algorithms using Python and OpenCV

The project aligns with Industry 4.0 principles, where intelligent systems are used to monitor and control manufacturing processes in real time. Unlike traditional sensor-based systems that detect only presence, this system uses computer vision to extract geometric and spatial information, enabling more accurate inspection.

1.2 TARGET CUSTOMERS

The target customers for this system are industries and organizations that rely on fastener-based assembly processes. These include:

- Automobile manufacturing industries, where bolt and nut assemblies are critical for structural integrity
- Small and Medium Enterprises (SMEs) involved in mechanical assembly operations
- Production units and fabrication workshops handling repetitive fastening tasks
- Quality control departments responsible for inspection and defect detection
- Educational and research institutions working on automation and inspection systems

In many of these environments, inspection is still carried out manually, especially in small-scale industries due to budget constraints. These customers require a solution that is:

- Affordable
- Reliable
- Easy to integrate
- Capable of real-time operation

The proposed system meets these requirements by providing a low-cost automated inspection solution using commonly available hardware and open-source software.

1.3 GO TO CUSTOMER (GTC) PROCESS

The Go-To-Customer (GTC) process was conducted to understand real-world problems faced by potential users and to validate the need for the proposed solution.

Customer Identification

Local workshops, fabrication units, and small-scale manufacturing industries were identified as primary stakeholders. These environments frequently use bolt–nut assemblies and require inspection processes.

Observation of Existing Practices

Field observation revealed that most industries rely on manual inspection methods, where operators visually check whether bolts and nuts are properly mounted. This process is:

- Time-consuming
- Dependent on operator skill
- Prone to fatigue-related errors

Interaction and Feedback Collection

Discussions with technicians and supervisors highlighted several concerns:

- Difficulty in maintaining consistency during long working hours
- Occasional oversight of missing or improperly mounted fasteners
- Need for faster inspection methods without increasing labor

Requirement Analysis

Based on the feedback, the following requirements were identified:

- Automated detection system with minimal human intervention
- Real-time inspection capability
- Cost-effective solution suitable for SMEs
- Non-contact inspection method to avoid mechanical wear

The insights obtained from the GTC process directly influenced the design and development of the proposed system.

1.4 POTENTIAL PROBLEMS IDENTIFIED

Based on customer interaction and field study, several problems were identified in the current inspection process:

1. Human Error

Manual inspection depends heavily on human attention and consistency. Over time, fatigue and repetitive tasks lead to errors such as:

- Missing detection of absent bolts
- Overlooking improper mounting

2. Inefficiency in Inspection

Manual inspection slows down the production process, especially in high-volume manufacturing environments.

3. Lack of Real-Time Monitoring

Existing systems do not provide immediate feedback, which may result in defective products moving further along the production line.

4. Limited Detection Capability

Conventional sensors can detect only the presence of an object but cannot verify:

- Proper alignment
- Correct assembly

5. High Cost of Industrial Systems

Advanced industrial inspection systems are expensive and not affordable for small-scale industries.

These problems indicate the need for a reliable, automated, and cost-effective inspection system.

1.5 SPECIFIC PROBLEM CHOSEN & RATIONALE – CUSTOMER FIT

Selected Problem

Detection of:

- Presence or absence of bolt and nut
- Verification of proper mounting and alignment

Technical Rationale

Fasteners play a critical role in maintaining the structural integrity of mechanical systems. Improper mounting can lead to:

- Mechanical failure
- Safety risks
- Reduced product lifespan

The selected problem is highly suitable for automation because:

- It is repetitive in nature
- Requires consistent inspection
- Can be solved using image processing techniques

Customer Fit Justification

The problem directly matches the needs identified during the GTC process. Customers require:

- Reduced manual effort
- Improved inspection accuracy
- Faster processing

The proposed solution fulfils these requirements by providing a vision-based inspection system that is both efficient and affordable.

1.6 FINAL PROBLEM STATEMENT

In manufacturing and assembly processes, ensuring the correct presence and proper mounting of bolt and nut assemblies is essential for maintaining product quality, safety, and reliability. However, traditional manual inspection methods are time-consuming, inconsistent, and prone to human error, especially in high-volume production environments.

There is a need for an automated inspection system that can perform real-time verification of fastener assemblies with high accuracy. The system should be cost-effective, easy to integrate with existing setups, and capable of detecting both missing and improperly mounted components.

Therefore, this project aims to develop a real-time bolt and nut mounting verification system using computer vision, where a sensor-controlled conveyor ensures proper positioning and image processing techniques are used for accurate detection and decision-making.

1.7 GTC AND PROBLEM – CUSTOMER FIT

The alignment between the identified problem and the proposed solution demonstrates a strong customer fit. The system addresses the key issues faced by customers:

- Reduces dependency on manual inspection
- Improves detection accuracy through algorithm-based analysis
- Provides real-time feedback during production
- Offers a cost-effective alternative to expensive industrial systems

From a technical standpoint, the system uses a hybrid architecture:

- Sensor for positioning and synchronization
- Vision system for detection and decision-making

This separation improves system reliability and performance. The proposed solution is scalable and can be adapted for different industrial applications, making it suitable for both small and large-scale environments.

Overall, the system provides a practical and efficient solution that meets customer requirements and enhances manufacturing quality standards.

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2. DESIGN AND DEVELOPMENT OF SOLUTION (STAGE – II)

2.1 SOLUTION PROCESS / METHODOLOGY

The proposed system is developed as a real-time vision-based inspection system integrated with a sensor-triggered conveyor mechanism for synchronized object positioning and analysis.

A conveyor system driven by a DC motor is used to transport test specimens (wooden blocks with bolt–nut assemblies) at a controlled speed. The spacing between successive objects is maintained to avoid overlapping during inspection. An IR proximity sensor is positioned at the inspection station to detect the arrival of each object.

Upon detection, the sensor generates a trigger signal to the control unit, which temporarily halts the conveyor. This ensures a stable inspection condition by eliminating motion-induced blur and positional variation.

A fixed-position camera is mounted above the inspection zone to capture images of the assembly. The acquired image is processed in real time using Python and the OpenCV library. The processing pipeline includes:

- Pre-processing: grayscale conversion and Gaussian filtering for noise reduction
- Segmentation: Region of Interest (ROI) extraction to isolate the fastener area
- Feature extraction: edge detection (Canny) and contour identification
- Geometric verification: evaluation of expected shape and positional alignment

A rule-based decision algorithm is implemented to verify:

- Presence of bolt head
- Presence of nut
- Relative alignment between bolt and nut

The output is classified as:

- OK – all features detected within defined thresholds
- NOT OK – missing or misaligned features

After processing, the control unit reactivates the conveyor, and the next object enters the inspection cycle. This closed-loop operation enables continuous, repeatable, and real-time inspection.

The mechanical layout of the conveyor-based inspection system was designed and validated using CAD modeling to ensure proper alignment, spacing, and structural stability of all components. The CAD model includes the conveyor frame, motor mounting arrangement, object path, IR sensor positioning, and camera mounting structure. This design ensures that each object is accurately positioned within the camera's field of view during inspection. Proper spacing between objects and precise sensor placement enable reliable triggering and synchronization between the mechanical and vision subsystems. The integration of CAD-based design with the implemented hardware setup improves system reliability, repeatability, and ease of assembly, making the overall solution suitable for practical deployment in automated inspection environments.



Figure 2.1(CAD model)

2.2 Minimum Viable Product (MVP)

The MVP is designed to validate the core functional requirements of the system with minimal hardware and algorithmic complexity.

Hardware Configuration:

- Conveyor mechanism with DC motor drive
- IR proximity sensor for triggering
- USB webcam for image acquisition
- Laptop/PC for processing

Software Configuration:

- Python programming environment
- OpenCV library for image processing

Functional Capabilities:

- Sensor-based object detection and conveyor control
- Synchronized image acquisition
- Basic feature-based detection of bolt and nut presence
- Real-time result visualization

The MVP demonstrates the feasibility of implementing a low-cost, vision-based inspection system without the need for expensive industrial vision hardware.

2.3 SOLUTION – CUSTOMER FIT

The proposed solution directly addresses key industrial challenges related to manual inspection inefficiency and quality inconsistency.

From a technical standpoint, the system improves inspection reliability by replacing subjective human judgment with deterministic image processing algorithms. The use of a vision-based approach enables detection of geometric features that cannot be captured using conventional sensors.

Key technical advantages include:

- Non-contact inspection, eliminating mechanical wear and sensor limitations
- High repeatability, due to algorithm-driven decision making
- Reduced cycle time, through real-time processing
- Low implementation cost, using standard imaging hardware and open-source tools

The architecture ensures optimal performance by separating system roles:

- Sensor → triggering and positional synchronization
- Vision module → feature extraction and decision logic

This makes the system suitable for small and medium-scale production environments, where cost and simplicity are critical constraints.

2.4 Concept MVP

The Concept MVP defines the functional architecture and data flow of the system.

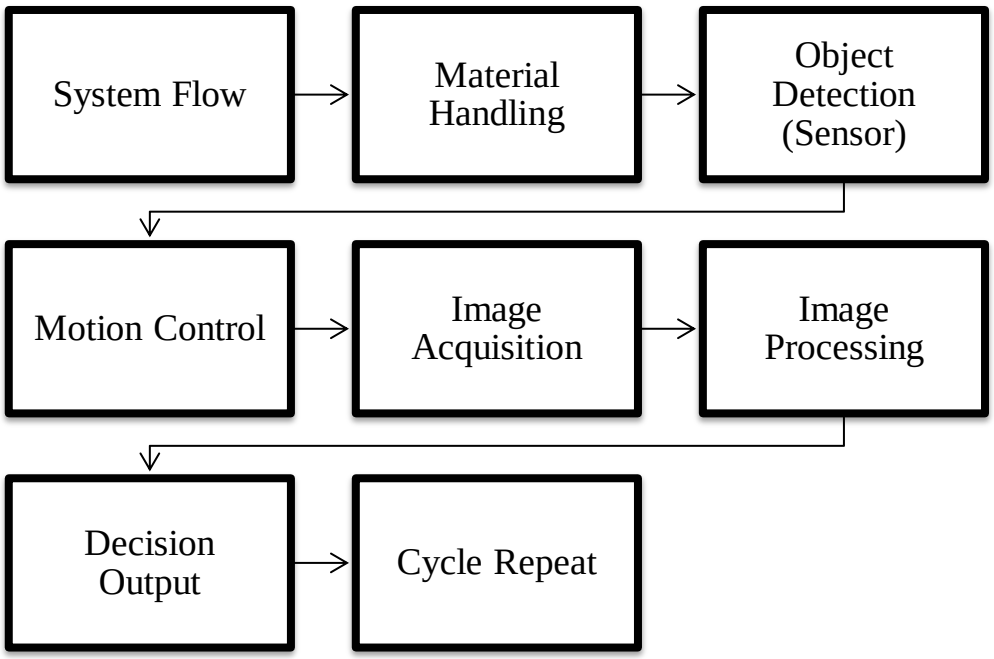


Chart 2.1(MVP)

Module Description:

- Input Module: Conveyor system and IR sensor
- Processing Module: Camera and image processing unit (OpenCV)
- Output Module: Display interface for result visualization

The system operates as a discrete inspection cycle, where each object is processed independently. This modular structure allows scalability and easy integration with advanced systems.

Future Scalability:

- Integration with PLC-based industrial control
- Multi-object detection using advanced algorithms
- AI/Deep Learning-based classification models
- Automated rejection mechanisms

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3. PROTOTYPE DEVELOPMENT (STAGE -III)

3.1 PROTOTYPE DEVELOPMENT

The prototype was developed to demonstrate a functional, real-time inspection system capable of verifying the presence and proper mounting of bolt–nut assemblies using a sensor-triggered conveyor and computer vision module.

3.1.1 Mechanical System Development

A mini conveyor system was designed and fabricated to simulate a production-line environment. The conveyor consists of:

- A rigid frame structure to support the system
- Rollers and a continuous belt mechanism
- A DC motor for driving the conveyor

The conveyor speed was selected to ensure:

- Smooth movement of objects
- Adequate time for sensor detection and system response
- Proper spacing between objects

Wooden blocks were used as carriers for bolt–nut assemblies. These blocks act as representative samples of real industrial components. Each block is placed on the conveyor with a fixed spacing to avoid overlapping during inspection.

The mechanical layout was designed using CAD tools to ensure:

- Proper alignment of components
- Stable platform for vibration-free operation
- Correct positioning of sensor and camera

3.1.2 Sensor and Control System

An Infrared (IR) proximity sensor was integrated into the system to detect the arrival of objects at the inspection station.

Working Principle:

- The sensor emits infrared radiation
- When an object comes within range, the reflected signal is detected
- A trigger signal is generated and sent to the control circuit

Control Action:

- The conveyor motor is temporarily stopped when the object is detected
- A fixed delay is introduced to allow image capture
- After processing, the motor resumes operation

This mechanism ensures:

- Accurate positioning of objects
- Elimination of motion blur
- Synchronization between hardware and software

The sensor is used strictly for triggering and positioning, not for defect detection.

3.1.3 Image Acquisition System

A webcam was mounted above the inspection zone to capture images of the bolt–nut assembly.

Design Considerations:

- Fixed mounting to maintain consistent field of view
- Adequate height to cover the inspection area
- Stable positioning to avoid vibration

Proper lighting conditions were ensured to:

- Improve image clarity
- Reduce shadows and reflections

- Enhance feature visibility

Images are captured only when the object is stationary, ensuring reliable input for processing.

3.1.4 Software Development and Processing

The detection system was implemented using Python programming language and the OpenCV library.

Processing Pipeline:

1. Image Capture
    - Frame captured from webcam
2. Pre-processing
    - Conversion to grayscale
    - Noise reduction using Gaussian filtering
3. Region of Interest (ROI) Selection
    - Isolation of the bolt–nut area
    - Reduction of background interference
4. Feature Extraction
    - Edge detection (Canny algorithm)
    - Contour detection to identify shapes
5. Verification Logic
    - Detection of bolt head
    - Detection of nut
    - Alignment verification based on position
6. Decision Output
    - OK → Properly mounted
    - NOT OK → Missing or improper mounting

The algorithm operates in real time, ensuring immediate feedback.

3.1.5 System Integration

All subsystems were integrated to function as a single unit:

- Conveyor → Object movement
- Sensor → Triggering and control
- Camera → Image acquisition
- Software → Processing and decision-making

The integration ensures a synchronized workflow, where each component operates in coordination with others.

3.1.6 Prototype Operation Cycle

The complete working cycle of the prototype is as follows:

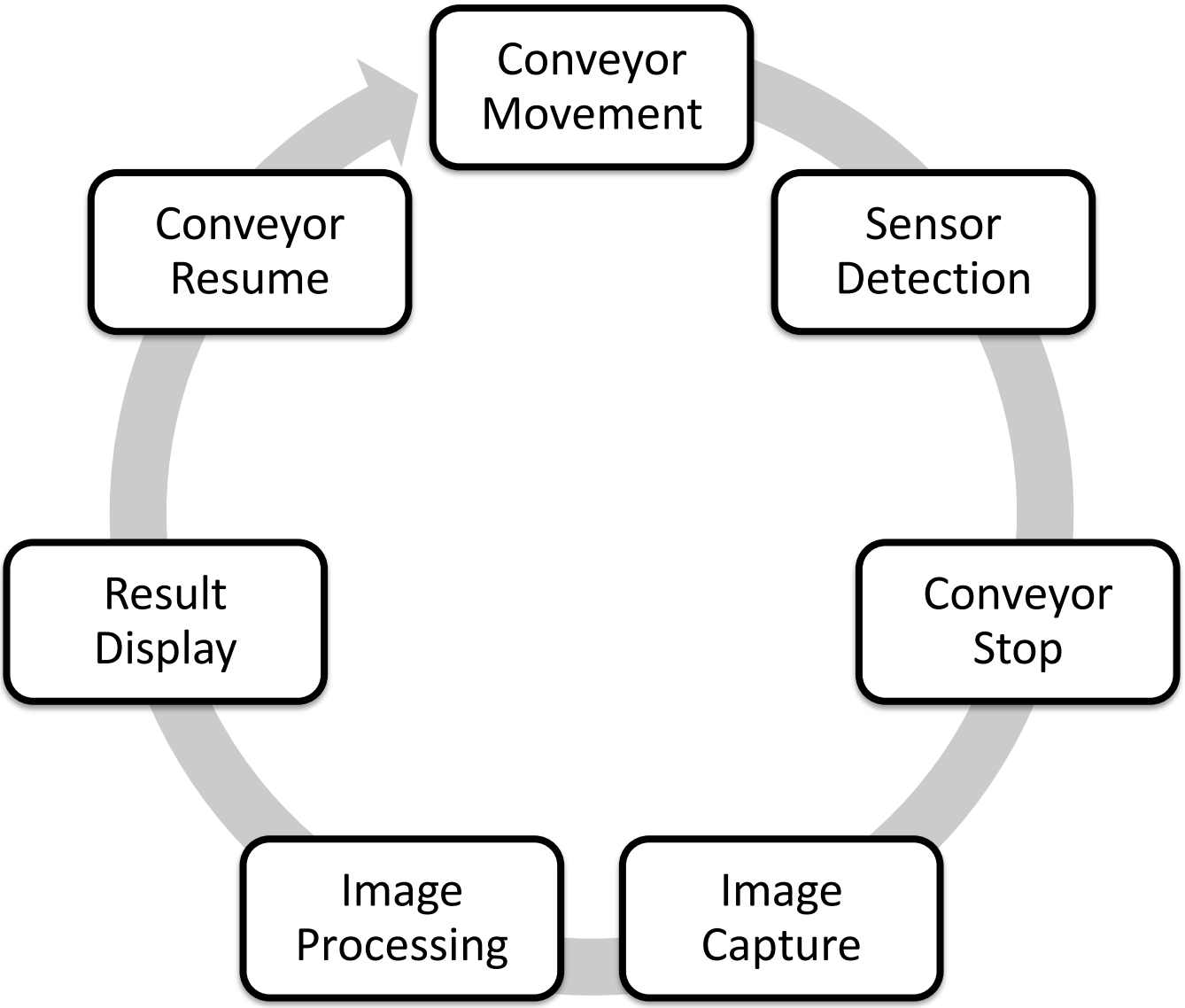


Chart 3.2(Prototype operation cycle)

Prototype image

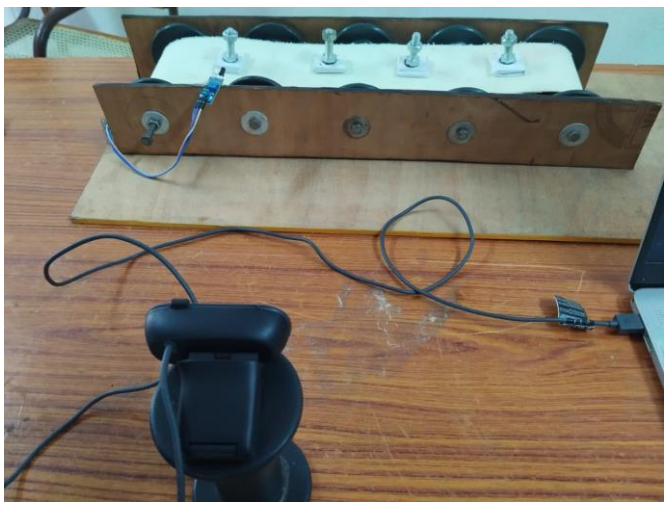


FIGURE 3.2(PROTOTYPE)

TESTING AND OUTPUT

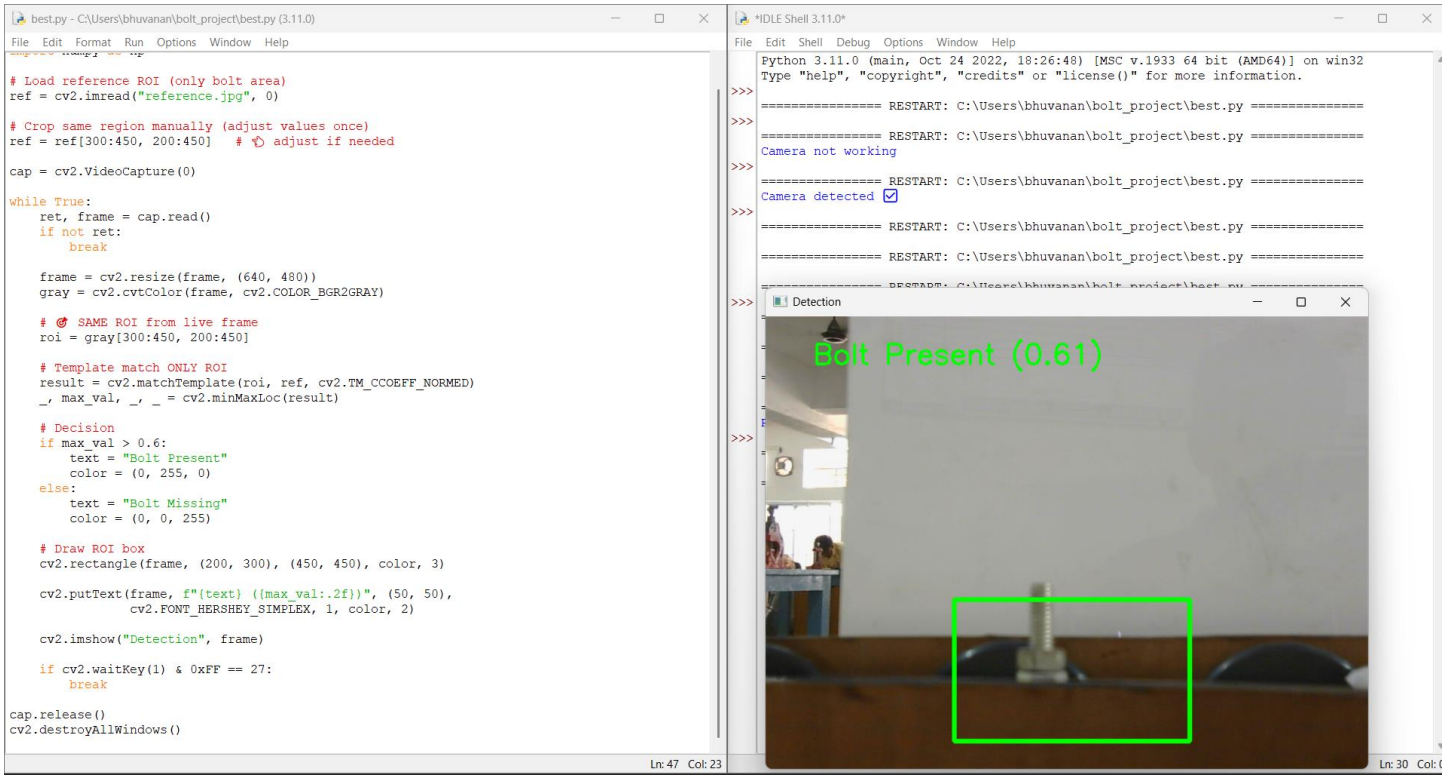


FIGURE 3.3(CODE RUNNING)

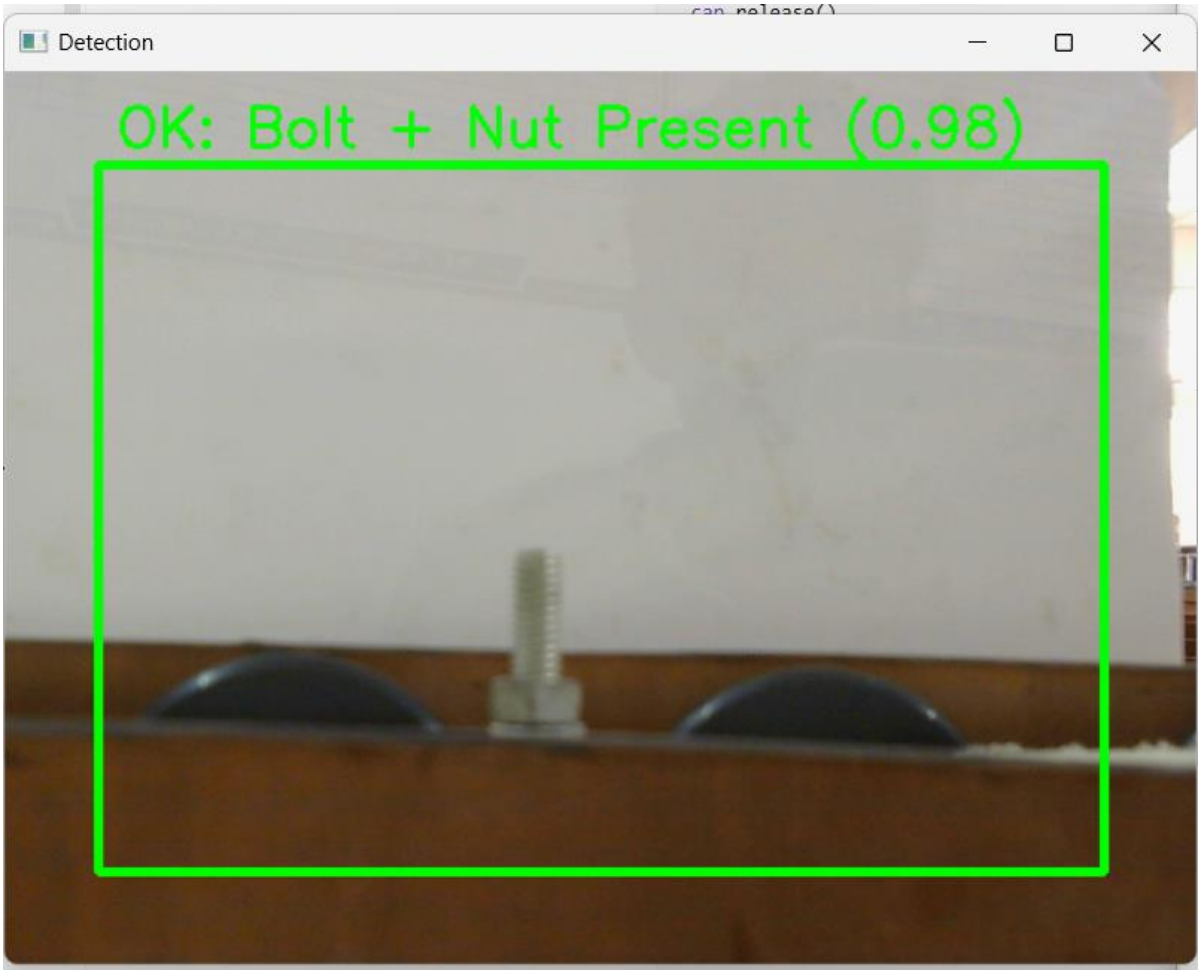


FIGURE 3.4(VERIFICATION PRESENT)

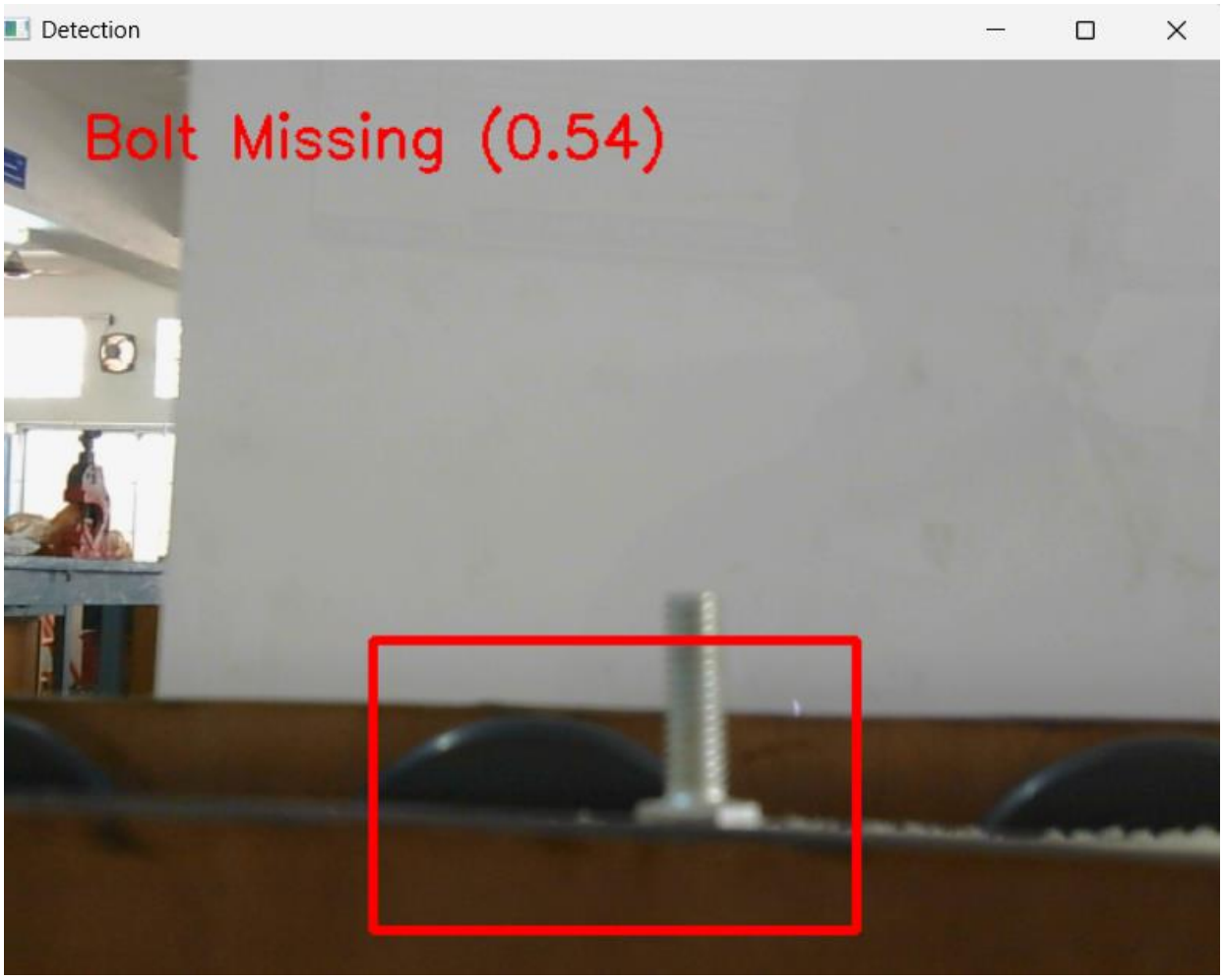


FIGURE 3.5(MISSING)

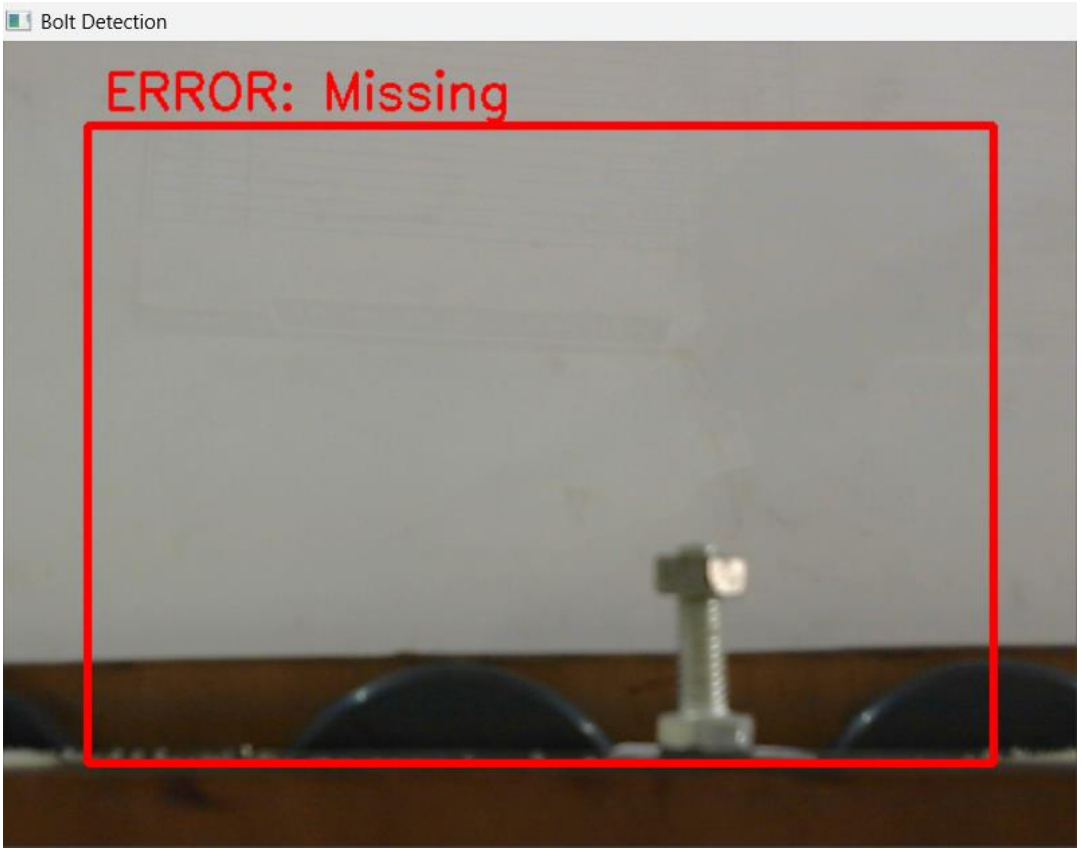


FIGURE 3.6(NOT PROPERLY ASSEMBLE)

3.2 GTC ITERATION & PRODUCT IMPROVEMENT

Based on feedback obtained during the Go-To-Customer (GTC) process and prototype testing, several improvements were identified and implemented.

3.2.1 Initial Observations

During early testing, the following issues were observed:

- Inconsistent detection due to lighting variations
- Slight misalignment of objects affecting accuracy
- Delay in synchronization between sensor and processing

3.2.2 Improvements Implemented

1. Lighting Optimization

- Improved illumination around inspection zone
- Reduced shadows and reflections

2. Camera Position Adjustment

- Fixed optimal height and angle
- Ensured consistent field of view

3. ROI Optimization

- Focused detection only on required region
- Reduced computational load

4. Timing Control

- Adjusted conveyor stopping duration
- Improved synchronization between hardware and software

5. Algorithm Refinement

- Improved contour detection thresholds
- Reduced false detections

3.2.3 Iterative Development Approach

The system was improved through multiple iterations:

1. Basic prototype testing
2. Identification of errors
3. Parameter tuning
4. Performance validation

This iterative process ensured that the final prototype achieved:

- Better accuracy
- Improved stability
- Reliable performance

3.3 PRODUCT – CUSTOMER FIT

The developed prototype was evaluated to determine how effectively it satisfies the requirements identified during the Go-To-Customer (GTC) analysis. The system demonstrates a strong alignment between customer needs and the implemented solution in terms of functionality, performance, and cost.

Functional Fit

The system successfully performs the required inspection tasks:

- Detection of the presence of bolt and nut
- Verification of proper mounting and alignment

The use of a vision-based approach ensures that the system can identify both missing and improperly mounted fasteners, which is not possible with conventional sensor-based methods alone. The real-time processing capability allows immediate feedback during operation.

Technical Fit

From a technical perspective, the system provides a robust and reliable solution:

- **Computer vision-based detection** enables accurate feature extraction and analysis
- **Sensor-based triggering** ensures proper positioning and synchronization
- **Non-contact inspection** avoids mechanical wear and increases durability
- **Real-time processing** improves inspection efficiency

The separation of functions enhances system performance:

- Sensor → positioning and control
- Vision system → detection and decision-making

Economic Fit

The system is designed to be cost-effective, making it suitable for small and medium-scale industries. The use of:

- Standard webcam for image acquisition
- Open-source tools such as Python and OpenCV
- Simple mechanical components

reduces overall implementation cost compared to industrial inspection systems.

Operational Fit

The system is easy to operate and integrate into existing workflows:

- Requires minimal training for operators
- Simple user interface for result display
- Compatible with conveyor-based production systems

The automated nature of the system reduces manual workload and improves consistency in inspection.

Scalability and Future Scope

The developed prototype provides a foundation for further enhancements, such as:

- Multi-object detection
- Integration with programmable logic controllers (PLC)
- AI-based advanced inspection models
- Automated rejection mechanisms

These improvements can extend the system for use in large-scale industrial environments.

4.BUSINESS MODEL DEVELOPMENT (STAGE -IV)

4.1 LEAN BUSINESS MODEL CANVAS (BMC)

The Lean Business Model Canvas provides a structured approach to analyze the commercial viability of the proposed system.

Customer Segments

- Small and Medium Manufacturing Enterprises (SMEs)
- Automobile assembly industries
- Fabrication and mechanical workshops
- Quality inspection units

These customers require low-cost automated inspection systems.

Value Proposition

- Real-time detection of bolt–nut mounting
- Reduction in human error
- Low-cost alternative to industrial vision systems
- Easy integration with existing conveyor setups
- Non-contact, vision-based inspection

Channels

- Direct industry visits and demonstrations
- Technical exhibitions and college projects showcase
- Online platforms and industrial networking
- Collaboration with local workshops

Customer Relationships

- Technical support and installation guidance
- Training for operators
- Regular updates and maintenance assistance

Revenue Streams

- Sale of complete inspection system
- Customization services for different industries
- Maintenance and upgrade services

Key Resources

- Hardware components (conveyor, sensor, camera)
- Software tools (Python, OpenCV)
- CAD design and prototype
- Technical knowledge and development team

Key Activities

- System design and development
- Prototype testing and validation
- Customer interaction and feedback collection
- Installation and support

Key Partners

- Hardware suppliers
- Local manufacturing units
- Academic institutions
- Automation solution providers

Cost Structure

- Component cost (motor, sensor, camera)
- Fabrication cost of conveyor
- Software development (minimal due to open-source tools)
- Maintenance and operational costs

4.2 MARKET POTENTIAL

The demand for automated inspection systems is increasing due to the rapid growth of manufacturing industries and the need for quality assurance.

Industry Demand

- Fastener inspection is critical in industries such as automotive, aerospace, and machinery manufacturing
- Increasing focus on quality control and zero-defect production

Target Market

- SMEs that cannot afford high-end industrial vision systems
- Local manufacturing units transitioning to automation

Market Opportunity

- Large number of small industries lacking automation
- Growing adoption of Industry 4.0 technologies
- Demand for cost-effective solutions

Competitive Advantage

Compared to existing systems:

- Lower cost
- Simpler design
- Easy implementation

4.3 BUSINESS PLAN AND MARKETING PLAN

Business Plan

The business model focuses on developing and supplying a low-cost automated inspection system.

Product Strategy

- Develop a modular inspection system
- Offer customization based on customer requirements

Pricing Strategy

- Affordable pricing for SMEs
- Lower cost compared to industrial systems

Revenue Model

- Direct product sales
- Customization and installation charges
- Maintenance services

Marketing Plan

Promotion Strategies

- Demonstration in local industries
- Technical presentations and exhibitions
- Online promotion through social media and websites

Distribution Strategy

- Direct sales to industries
- Collaboration with automation solution providers

Customer Acquisition

- Target small-scale industries first
- Build trust through prototype demonstration
- Use feedback to improve product

4.4 BUSINESS – CUSTOMER FIT

The proposed system strongly aligns with customer requirements.

Customer Needs Addressed

- Reduction in manual inspection
- Improved accuracy and consistency
- Real-time defect detection
- Affordable automation

Solution Alignment

- Vision-based system provides accurate detection
- Sensor ensures proper positioning
- Real-time processing improves productivity

Benefits to Customers

- Reduced labor cost
- Improved product quality
- Faster inspection process
- Easy integration

4.5 PATENTING POTENTIAL AND STARTUP POTENTIAL

Patenting Potential

The system has potential for patenting in areas such as:

- Integration of sensor-triggered conveyor with vision system
- Low-cost inspection mechanism design
- Specific algorithm for fastener detection

However, further innovation and uniqueness in algorithm or hardware design may be required for strong patent claims.

Startup Potential

The project has strong potential to be developed into a startup:

- High demand for low-cost automation
- Scalable design
- Applicable across multiple industries

Startup Opportunities

- Manufacturing and selling inspection systems
- Providing automation solutions for SMEs
- Offering customization services

4.6 SDG ALIGNMENT

The project aligns with the United Nations Sustainable Development Goals (SDGs):

SDG 9: Industry, Innovation, and Infrastructure

- Promotes industrial automation
- Encourages innovation in manufacturing

SDG 8: Decent Work and Economic Growth

- Improves productivity
- Supports industrial growth

SDG 12: Responsible Consumption and Production

- Reduces defective products
- Improves resource efficiency

4.7 UNIQUE SELLING PROPOSITION (USP)

The Unique Selling Proposition of the system is:

- Low-cost alternative to expensive industrial inspection systems
- Real-time vision-based detection of bolt–nut mounting
- Simple and modular design for easy implementation
- Sensor + vision integration for reliable performance

Key Differentiators

- Affordable for small industries
- Easy to operate and maintain
- Scalable for future upgrades

5. SUMMARY (STAGE V)

5.1 MINIMUM VIABLE PRODUCT (MVP)

The Minimum Viable Product (MVP) of the proposed system represents a functional prototype designed to validate the feasibility of a real-time bolt and nut mounting verification system. The MVP integrates essential mechanical, electronic, and software components to perform automated inspection.

The system consists of a conveyor mechanism driven by a DC motor to transport test objects, an IR proximity sensor to detect object arrival and control conveyor motion, and a camera-based vision system for inspection. The image captured by the camera is processed using Python and OpenCV, enabling detection of bolt and nut presence as well as verification of proper mounting.

The MVP demonstrates key functionalities such as:

- Sensor-based triggering and conveyor control
- Stable positioning of objects for inspection
- Real-time image acquisition
- Feature-based detection of bolt and nut
- Display of inspection results (OK / NOT OK)

The prototype successfully validates that a low-cost vision-based system can perform inspection tasks typically handled by manual processes or expensive industrial systems.

5.2 USP (TECHNICAL DIMENSION)

From a technical perspective, the proposed system offers several unique advantages that distinguish it from conventional inspection methods.

The primary USP lies in the use of computer vision for fastener verification, enabling detection of both presence and proper mounting, which cannot be achieved using traditional sensors alone. The system employs image processing techniques such as edge detection, contour analysis, and region-based processing to extract relevant features from captured images.

Another important technical feature is the separation of system functions:

- The IR sensor is used only for triggering and positioning
- The vision system performs detection and decision-making

This architecture improves system reliability and reduces dependency on hardware-based detection.

The system also operates in real time, ensuring that inspection results are generated immediately during the production process. Additionally, the use of non-contact inspection eliminates wear and tear associated with physical sensors.

Overall, the system provides a scalable, modular, and efficient inspection solution that can be extended with advanced algorithms in the future.

5.3 USP (BUSINESS DIMENSION)

From a business standpoint, the proposed system offers a strong value proposition, particularly for small and medium-scale industries.

The key USP is its cost-effectiveness, as the system utilizes:

- Standard webcam for image acquisition
- Open-source software tools such as Python and OpenCV
- Simple mechanical components

This significantly reduces the overall cost compared to industrial vision systems, making it accessible to a wider range of customers.

The system also provides:

- Reduced labor dependency
- Improved inspection accuracy
- Increased production efficiency

Its ease of implementation and operation further enhances its market potential. Industries can adopt the system without major modifications to existing setups, making it a practical and attractive solution.

5.4 INNOVATIVENESS & IP POTENTIAL

The project demonstrates innovation through the integration of low-cost hardware with computer vision techniques to solve a real industrial problem. While vision-based inspection systems exist, the novelty of this project lies in:

- Combining a sensor-triggered conveyor system with vision-based detection
- Developing a cost-effective alternative to expensive industrial solutions
- Implementing a simple yet effective algorithm for fastener verification

From an Intellectual Property (IP) perspective, the system has potential in areas such as:

- System architecture design
- Integration methodology
- Application-specific inspection algorithms

Further development and refinement, such as introducing AI-based detection or advanced automation features, can enhance the patentability of the system.

5.5 SCOPE FOR STARTUP CREATIONS

The project has strong potential to be developed into a startup focused on industrial automation and inspection solutions.

The increasing demand for automation in manufacturing industries creates a favorable environment for such a venture. The system can be offered as:

- A standalone inspection product
- A customized solution for specific industries
- A service-based model including installation and maintenance

The scalability of the system allows expansion into:

- Multi-component inspection systems
- AI-based quality control solutions
- Integration with industrial automation platforms

With proper development and market strategy, the project can evolve into a commercially viable product and startup opportunity.

5.6 FINAL MINIMUM VIABLE PRODUCT (MVP)

The final MVP represents the improved version of the prototype after incorporating feedback and iterative enhancements.

The system achieves:

- Stable conveyor operation with synchronized control
- Accurate positioning using sensor-based triggering
- Reliable image acquisition with consistent quality
- Improved detection accuracy through refined algorithms

- Real-time result display with minimal delay

The final MVP demonstrates a fully functional, integrated system capable of performing automated inspection of bolt and nut assemblies under controlled conditions.

It serves as a strong foundation for further development, including:

- Industrial-scale implementation
- Advanced feature integration
- Commercial product development

6.KEY LEARNING OF THE TEAM (10 POINTS) (STAGE -VI)

1. Integration of Mechanical and Software Systems
   The team learned how to integrate a mechanical conveyor system with software-based image processing, understanding the importance of synchronization between hardware and software components.

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2. Practical Application of Computer Vision
   The project provided hands-on experience in using Python and OpenCV for real-time image processing, including techniques such as edge detection, contour analysis, and object verification.

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3. Importance of System Synchronization
   The team understood that accurate detection depends on proper timing between sensor triggering, conveyor stopping, and image capture.

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4. Role of Sensors in Automation
   It was learned that sensors are not always used for detection but can be effectively used for triggering and positioning control in automated systems.

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5. Design Considerations in Mechanical Systems
   The team gained knowledge in designing a stable conveyor system, maintaining proper spacing between objects, and ensuring vibration-free operation for accurate inspection.

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6. Effect of Lighting and Environment on Detection Accuracy
   The project highlighted the impact of lighting conditions, shadows, and reflections on image processing accuracy and the need for controlled illumination.

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7. Development of Real-Time Processing Systems
   The team learned how to implement a system that performs continuous inspection with minimal delay, ensuring real-time operation.

8. Problem Identification through Customer Interaction (GTC)

   Through the Go-To-Customer process, the team gained experience in identifying real industrial problems and designing solutions based on user requirements.

9. Iterative Development and Optimization

   The project emphasized the importance of testing, identifying issues, and improving the system through multiple iterations to achieve better performance.

10. Understanding of Industrial Automation Concepts

    The team developed a clear understanding of automation systems, including material handling, inspection processes, and quality control mechanisms used in industries.

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